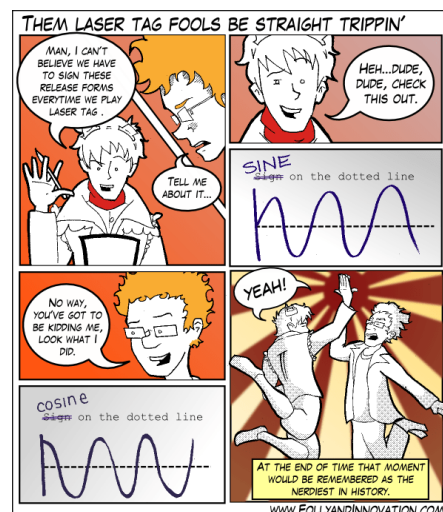


Unit 6: Trigonometric Functions

Expectations

By the end of this unit, students will be able to:

- Describe key properties of periodic functions arising from real-world applications;
- Represent a sinusoidal function with an equation, given its graph or its properties;
- Sketch the graphs of sine and cosine functions and described their key properties (domain, range, amplitude, axis of the curve, max/min values);
- Apply parameters a , k , d , and c to sketch transformed sine and cosine functions.



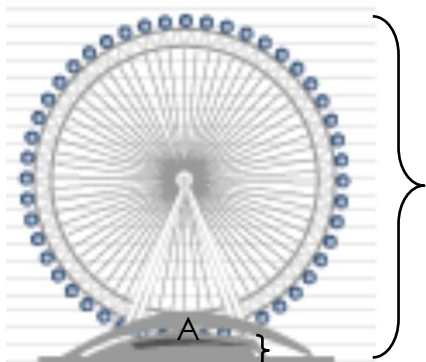
Homework Log

Lesson	Date	Section	Assigned Work	Extra Practice
1		6.1 Periodic Functions	Text: p. 352 #1 – 5, 7 – 11	
2		Graphing the Sine and Cosine Parent Functions	Text: p. 370 #1 – 6b, 8a, 13 – 14	
3		Transformations of Sinusoidal Functions Part 1	Text: p. 379 #1 – 3	Text: pg. 376 #4, 5
4		Transformations of Sinusoidal Functions Part 2	Text: pg. 383 #4 – 6, 9 – 12	Text: pg. 383 #1 – 3, 7
5		Determining the Equation of Sinusoidal Functions	Text: pg. 391 #3, 4 (sin & cos for each), 6, 7, 11, 13, 14	Text: pg. 391 #1 (sin & cos for each), 2, 5, 9, 10
6		Applications of Sinusoidal Functions	Text: pg. 398 #1 – 10, 13	
7		Review for Test	LP: pg. 20 – 23	Text: pg. 404 #2 – 4, 6, 8 – 10, 12, 13

Modelling Periodic Behaviour

The world's tallest Ferris Wheel, the High Roller, opened in Las Vegas, Nevada on March 31, 2014. The 520-foot diameter wheel, which has its centre 290 feet above the ground, will give up to 1120 passengers per hour a fantastic view of the city and surrounding area. Each of the wheel's 32 capsules holds 40 people.

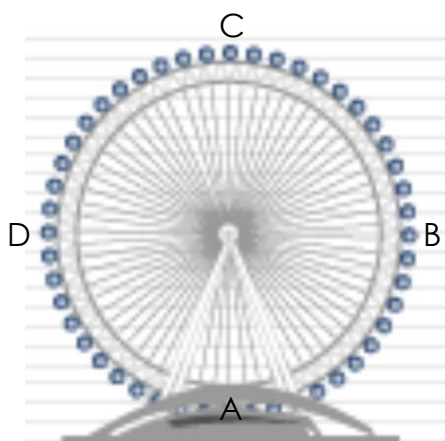
State the maximum and minimum height of the High Roller.



Suppose that you and a group of friends are riding the Ferris wheel. As shown in the diagram, you enter a gondola at point A. The High Roller turns counter clockwise at a constant speed and takes 30 minutes to complete one revolution.

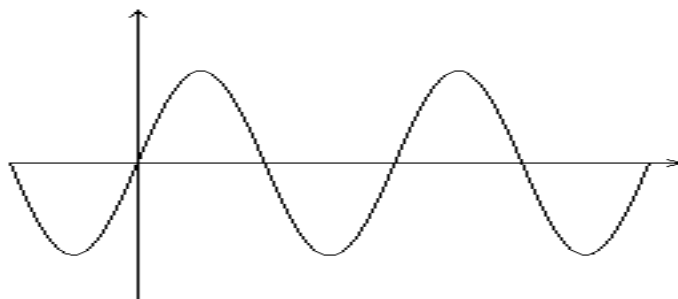
Identify the coordinates, in terms of time (minutes) and height (feet) of the points B, C, and D. Plot the points on a grid. Sketch a curve of best fit to show the relationship between your height, h , relative to the x-axis, and the time, t . Think in terms of a ball of string unravelling, creating a wave.

Time (minutes)					
Height (feet)					



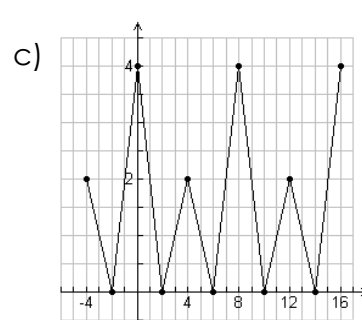
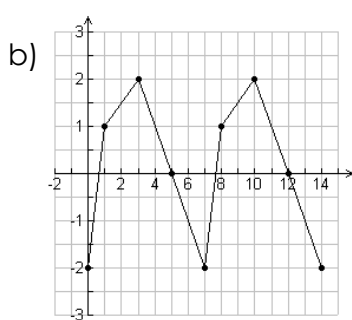
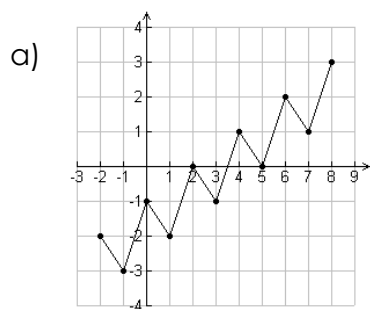
Key Ideas

- A function is **periodic** if it has a pattern of y-values that repeats at regular intervals.
- One complete pattern is called a **cycle**.
- The horizontal length of one cycle is called the **period** of the function.
- The horizontal line that is halfway between the maximum and minimum values of a periodic curve is called the **axis of the curve**.
- The equation of the axis of the curve is $y = \frac{\text{maximum value} + \text{minimum value}}{2}$
- The magnitude of the vertical distance from the axis of the curve to either the maximum (**peak**) or minimum (**trough**) value is called the **amplitude** of the function.
- The **amplitude**, a , is calculated as $a = \frac{\text{maximum value} - \text{minimum value}}{2}$

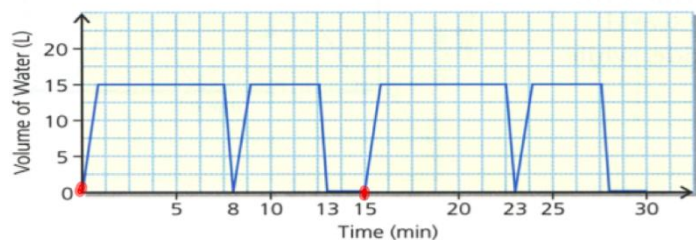


Note: A cycle may begin at any point of the graph (One cycle = One period)

Example 1: Determine if the function is periodic. If so, state the period, the maximum value, the minimum value, the amplitude and the equation of the axis of the curve.

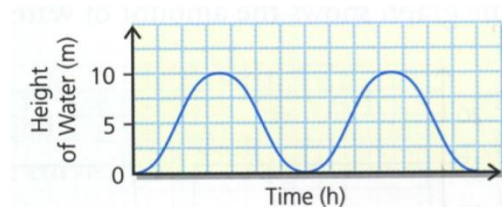


Example 2: The automatic dishwasher in a school cafeteria runs constantly through lunch. The graph shows the amount of water used as a function of time.



- Explain why the operation of the dishwasher is an example of a periodic function.
- What is the length of the period? What does one complete cycle mean in the context of the question?
- Extend the graph for one more complete cycle.

Example 3: The Bay of Fundy, which is between New Brunswick and Nova Scotia, has the highest tides in the world. There can be no water on the beach at low tide, while at high tide the water covers the beach.

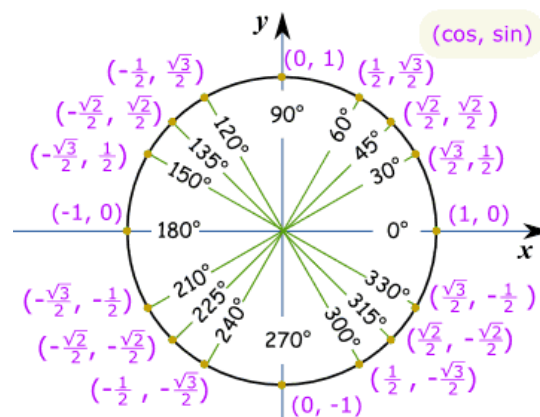


- Why can you use the periodic functions to model the tides?
- What is the change in depth of water from low tide to high tide?
- Determine the equation of the axis of the curve.
- What is the amplitude of the curve?

Investigation: Sketching the Graphs of $y = \sin \theta$ and $y = \cos \theta$

Recall: The Unit Circle

A circle with a radius of one unit is known as 'the unit circle'. When a unit circle is constructed with centre (0,0) it is very useful in establishing trigonometric ratios for special angles.

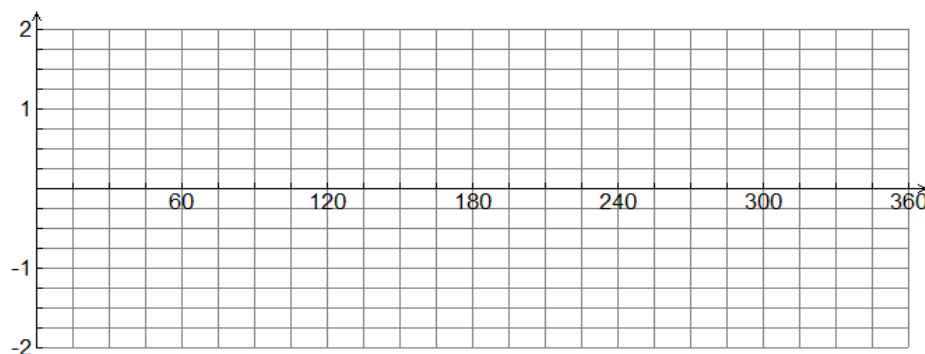


Part A: The Graph of $y = \sin \theta$

Complete the following table for the values of $y = \sin \theta$, $0^\circ \leq \theta \leq 360^\circ$. Include the exact value of $\sin \theta$, and the decimal value for $\sin \theta$, rounded to the nearest tenth.

Value of θ	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
Exact value of $\sin \theta$													
Decimal value of $\sin \theta$													

Plot the ordered pairs $(\theta, \sin \theta)$ on the grid below. Draw a curve of best fit to join the points.



1. What is the maximum value of $y = \sin \theta$? For what value of θ does the maximum value of y occur?
2. What is the minimum value of $y = \sin \theta$? For what value of θ does the minimum value of y occur?
3. What is the amplitude of $y = \sin \theta$?

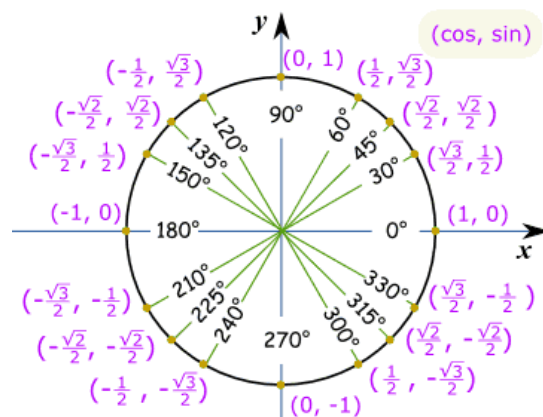
4. Explain whether the graph of $y = \sin \theta$ is periodic. If so, what is the period?

5. Explain whether the graph of $y = \sin \theta$ is a function.

6. For the function $y = \sin \theta$, state:

a) the domain

b) the range

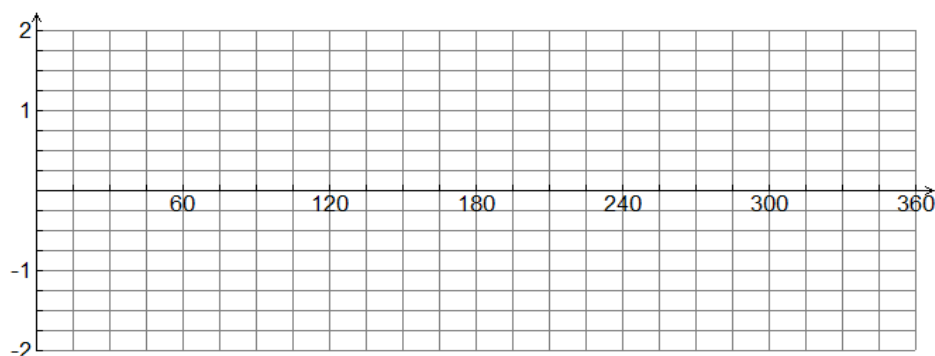


Part B: The Graph of $y = \cos \theta$

Complete the following table for the values of $y = \cos \theta$, $0^\circ \leq \theta \leq 360^\circ$. Include the exact value of $\cos \theta$, and the decimal value for $\cos \theta$, rounded to the nearest tenth.

Value of θ	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
Exact value of $\cos \theta$													
Decimal value of $\cos \theta$													

Plot the ordered pairs $(\theta, \cos \theta)$ on the grid below. Draw a curve of best fit to join the points.



1. What is the maximum value of $y = \cos \theta$? For what value of θ does the maximum value of y occur?
2. What is the minimum value of $y = \cos \theta$? For what value of θ does the minimum value of y occur?

3. What is the amplitude of $y = \cos \theta$?
4. Explain whether the graph of $y = \cos \theta$ is periodic. If so, what is the period?
5. Explain whether the graph of $y = \cos \theta$ is a function.
6. For the function $y = \cos \theta$, state:
 - a) the domain
 - b) the range
7. Compare the graphs of $y = \sin \theta$ and $y = \cos \theta$. How are they the same? How are they different?

Transformations of Sinusoidal Functions Part 1

Vertical Stretches/Compressions: $y = a\sin x$ and $y = a\cos x$

Mapping Notation: $(x, y) \rightarrow (x, ay)$

Example 1: Graph $y = 2\sin x$, and $y = \frac{1}{2}\cos x$, for $0^\circ \leq x \leq 360^\circ$.

$y = \sin x$	
x	y
0°	
90°	
180°	
270°	
360°	

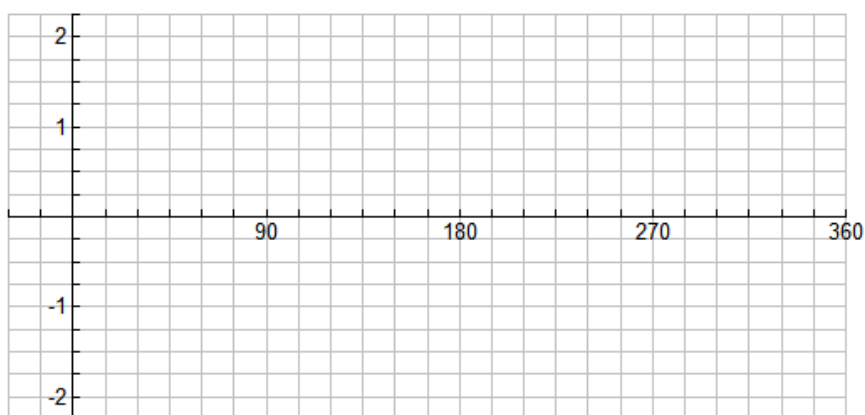
 $\Rightarrow$

$y = 2\sin x$	
x	y

$y = \cos x$	
θ	y
0°	
90°	
180°	
270°	
360°	

 $\Rightarrow$

$y = \frac{1}{2}\cos x$	
x	y



- What points are affected?
- What points are unaffected (invariant)?
- What is the amplitude, a , of the function?
- What is the period?
- What is the equation of the axis of the curve?
- State the domain and range.

$y = 2\sin x$

$y = \frac{1}{2}\cos x$

Generally,

- for $a > 1$, the graph is stretched vertically (expanded) by a factor of a .
- for $0 < a < 1$, the graph is compressed vertically by a factor of a .

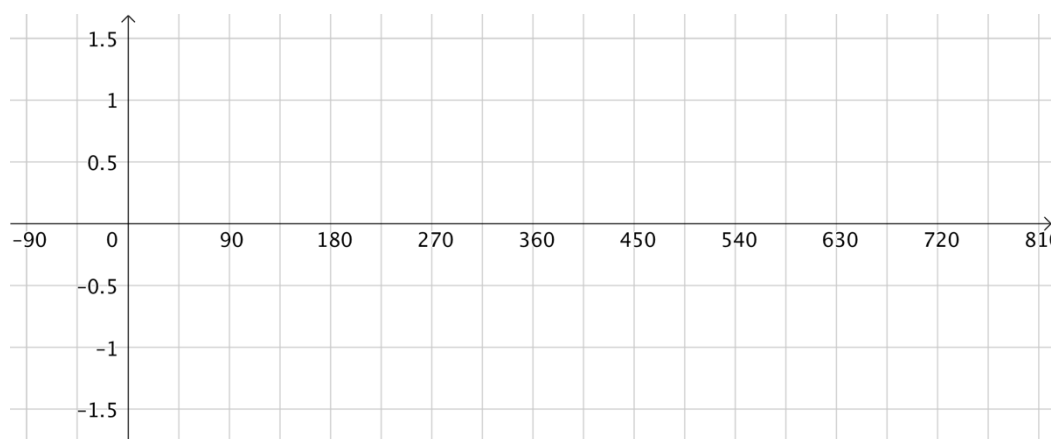
The amplitude of each function $y = \sin x$ and $y = \cos x$ is a .

Horizontal Stretches/Compressions: $y = \sin(kx)$ and $y = \cos(kx)$

Mapping Notation: $(x, y) \rightarrow \left(\frac{1}{k}x, y\right)$

Example 2: Graph one cycle of $y = \cos(2x)$ and $y = \sin\left(\frac{1}{2}x\right)$.

$y = \cos x$		$\Rightarrow$	$y = \cos(2x)$			$y = \sin x$		$\Rightarrow$	$y = \sin\left(\frac{1}{2}x\right)$	
x	y		x	y		x	y		x	y
0°						0°				
90°						90°				
180°						180°				
270°						270°				
360°						360°				



- What points are affected?
- What points are unaffected (invariant)?
- What is the amplitude, a , of the function?
- What is the period?
- What is the equation of the axis of the curve?
- State the domain and range.

$$y = \cos(2x)$$

$$y = \sin\left(\frac{1}{2}x\right)$$

_____	_____
_____	_____
_____	_____
_____	_____
_____	_____

Generally,

- for $k > 1$, the graph is horizontally compressed by a factor of k .
- for $0 < k < 1$, the graph is horizontally stretched (expanded) by a factor of k .

The value of k determines the number of degrees in the period of the graph. To determine the period of the trigonometric function, divide the period of the base curve by k .

For example, $y = \sin 3x$ has period: _____ $y = \cos\left(\frac{1}{3}x\right)$ has period: _____

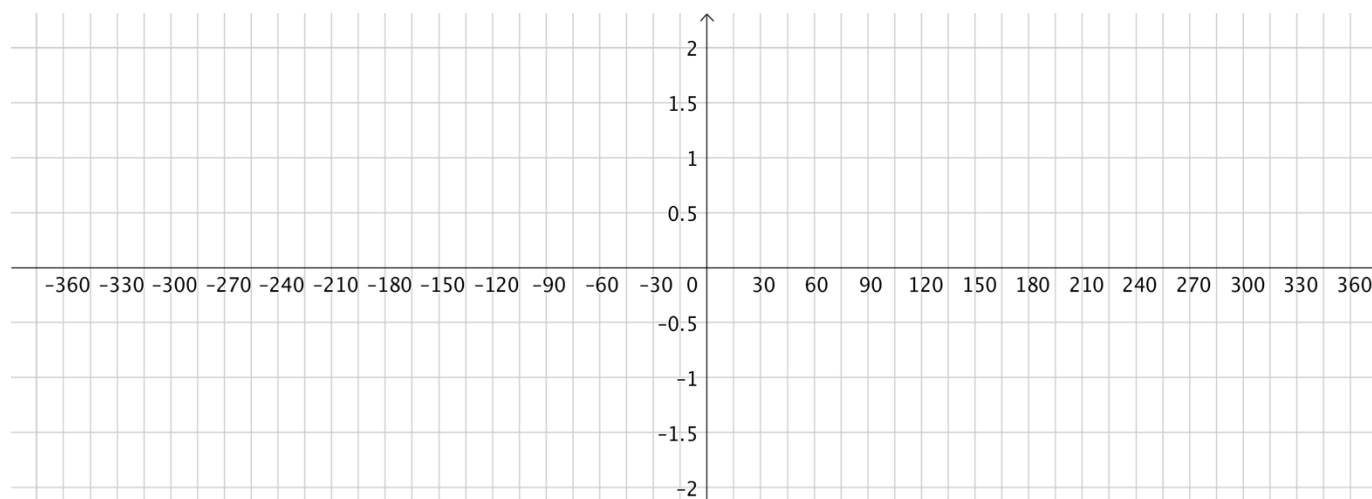
Vertical/Horizontal Reflections:**Recall:**

- $y = -f(x)$ reflects the function across the x-axis.
- $y = f(-x)$ reflects the function across the y-axis.

Mapping Notation: $(x, y) \rightarrow (-x, -y)$

Example 3: Graph $y = -\sin x$, and $y = \cos(-x)$.

$y = \sin x$		$\Rightarrow$	$y = -\sin x$		$y = \cos x$	$\Rightarrow$	$y = \cos(-x)$	
x	y		x	y	x	y	x	y
0°					0°			
90°					90°			
180°					180°			
270°					270°			
360°					360°			



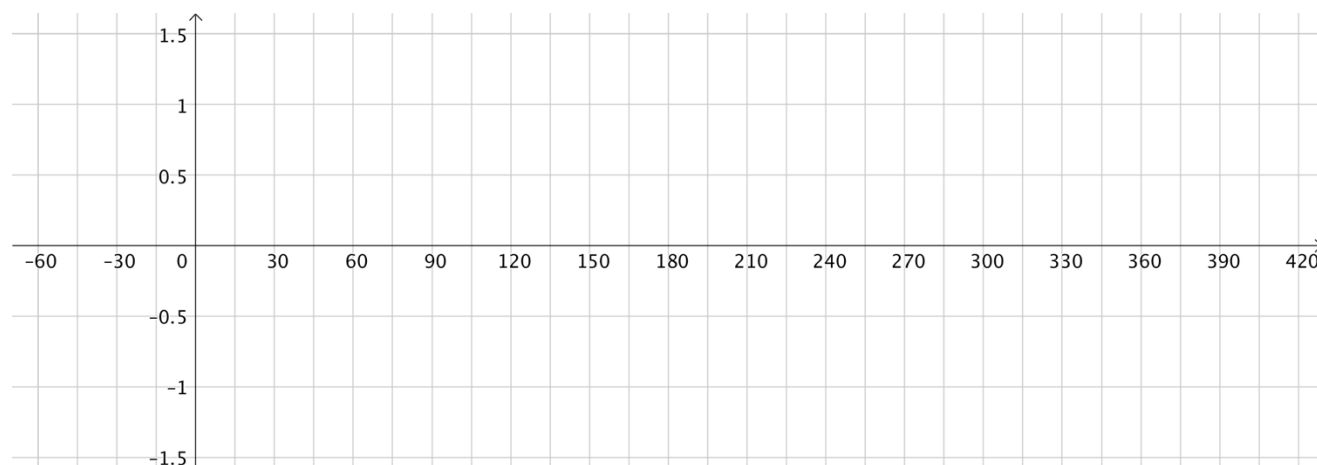
State its period, amplitude, axis of the curve, domain and range of each transformed function.

Horizontal Translations/Phase Shifts: $y = \sin(x \pm d)$ and $y = \cos(x \pm d)$

Mapping Notation: $(x, y) \rightarrow (x \pm d, y)$

Example 4: Graph $y = \sin(q + 30^\circ)$.

$y = \sin x$		$\Rightarrow$	$y = \sin(x + 30^\circ)$	
x	y		x	y
0°				
90°				
180°				
270°				
360°				



State the period, amplitude, axis of the curve, domain and range of the transformed function.

Vertical Translations: $y = \sin x \pm c$ and $y = \cos x \pm c$

Recall: $(x, y) \rightarrow (x, y \pm c)$

Example 5: Graph $y = \cos \theta + 3$.

$y = \cos x$		$\Rightarrow$	$y = \cos x + 3$	
x	y		x	y
0°				
90°				
180°				
270°				
360°				



State the period, amplitude, axis of the curve, domain and range of the transformed function.

Transformations of Sinusoidal Functions Part 2

$$g(x) = a \sin(k(x - d)) + c \text{ and } g(x) = a \cos(k(x - d)) + c$$

Ex 1: Refer to $y = 2 \sin(2(x - 90^\circ)) + 4$ for the following questions.

a) State the amplitude, the period and the equation of the axis of the curve.

b) Graph the function. State the domain and range for one cycle.



Ex 2: Refer to $y = 3 \cos(2(x + 45^\circ)) - 1$ for the following questions.

a) State the amplitude, the period and the equation of the axis of the curve.

b) Graph the function. State the domain and range for one cycle.



Ex 3: Refer to $y = -2\cos\left(\frac{1}{3}x - 30^\circ\right) + 3$ for the following questions.

a) State the amplitude, the period and the equation of the axis of the curve.

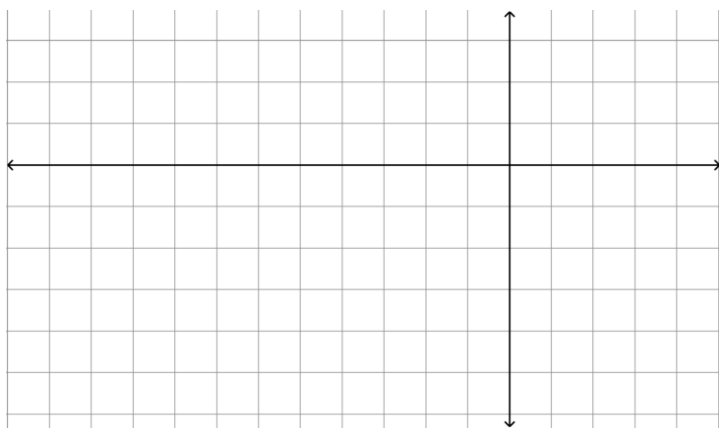
b) Graph the function. State the domain and range for one cycle.



Ex 4: Refer to $y = \frac{1}{2}\sin(-3x + 180^\circ) - 1$ for the following questions.

a) State the amplitude, the period and the equation of the axis of the curve.

b) Graph the function. State the domain and range for one cycle.



Determining the Equation of a Trigonometric Function

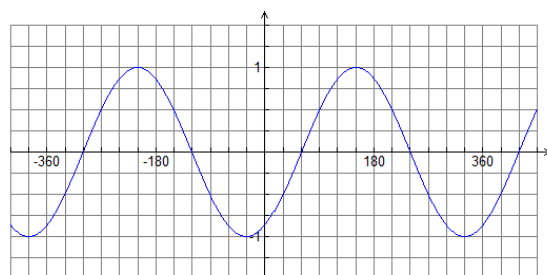
If you are given a set of data and the corresponding graph is a sinusoidal function, then you can determine the equation by calculating the graph's period, amplitude, and equation of the axis. This information will help you determine the values of k , a , and c , respectively, in the equations $g(x) = a \sin(k(x - d)) + c$ and $g(x) = a \cos(k(x - d)) + c$. The value of d is determined by estimating the required horizontal shift (left or right) compared with the graph of the sine or cosine curve.

Need to Know

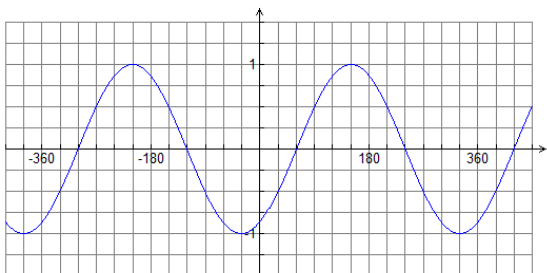
- If the graph begins at a maximum value, it may be easier to use the cosine function as your model.
- The domain and range of a sinusoidal model may need to be restricted for the situation you are dealing with.

Example 1: A sinusoidal function is graphed below.

- a) The function can be considered as a sine function. Determine two possible values for the phase shift. What is the equation of the function for each phase shift?

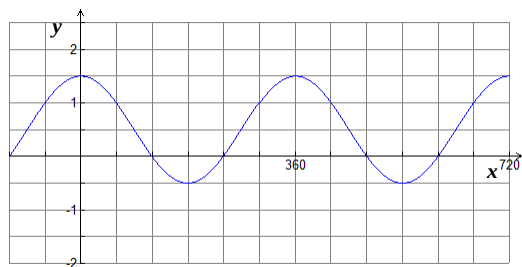


- b) The function can be considered as a cosine function. Determine two possible values for the phase shift. What is the equation of the function for each phase shift?

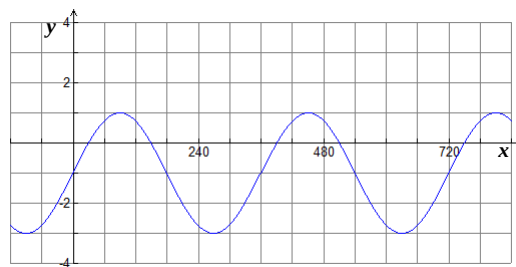


Example 2: Each function below is sinusoidal. Write two equations to represent each function.

a)

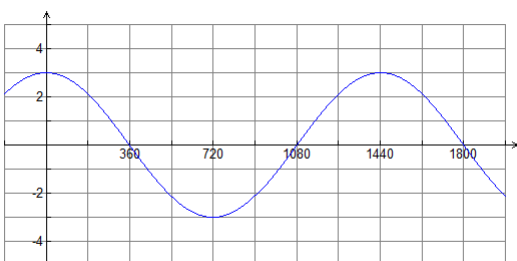


b)

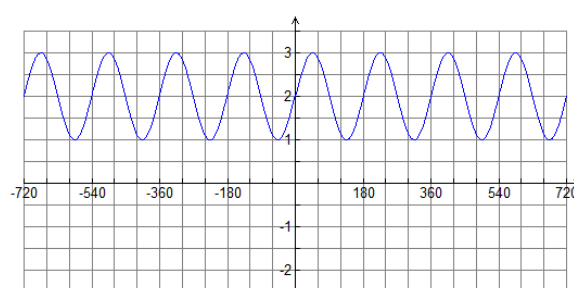


Example 3: Determine a sine and cosine equation for each function.

a)



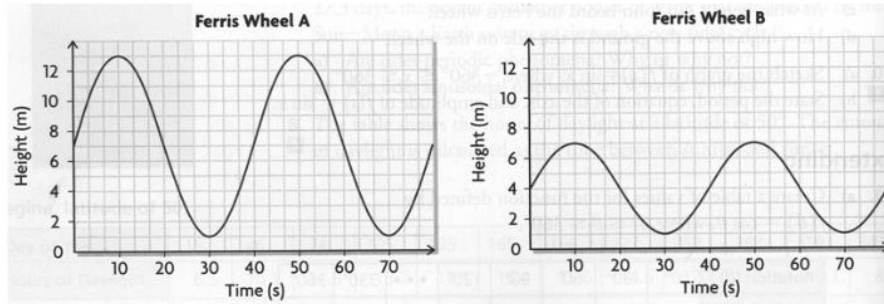
b)



Example 4: A sinusoidal function has an amplitude of 2 units, a period of 180° , and a maximum point at $(0, 3)$. Represent the function with an equation in two different ways.

Applications of Sinusoidal Functions

Example 1: At an amusement park, Ms. Vena had different students ride two Ferris wheels. John rode on Ferris wheel A, and Laura rode on Ferris wheel B. Ms. Vena collected data and produced two graphs.



What is the period of each function, and what does it represent?

Graph A

Graph B

What is the equation of each axis of the curve, and what does it represent?

Graph A

Graph B

What is the amplitude of each function, and what does it represent?

Graph A

Graph B

State an equation that represents each function.

Graph A

Graph B

Who is travelling faster, John or Laura?

Comparing Speeds in Sinusoidal Functions

Recall:

Graph A

Graph B

Reflecting

1. How does changing the radius of the wheel affect the sinusoidal graph?
2. How does changing the height of the axle of the wheel affect the sinusoidal graph?

Example 2: The top of a flagpole sways back and forth in high winds. The top sways 10 cm to the right (+10 cm) and 10 cm to the left (-10 cm) of its resting position and moves back and forth 4 times per second. At $t = 0$, the pole was momentarily at its resting position. Then it starts moving to the right.

- a) Draw a graph to describe the distance the top of the pole is from its resting position in terms of time.

b) Find an equation for the graph.

c) Determine the position of the pole after 1.3 s.

d) If the wind speed decreases such that the pole moves back and forth only 2 times per second, what would be the new equation of the sinusoidal function?

Example 3: In summer, the sun sets later than it does in winter. The time of the sunset can be important information for some people including parents, environmentalists, and film-makers. The latest sunsets occur around June 21 (the 172nd day of the year), and the earliest ones occur around December 21 (the 355th day of the year).

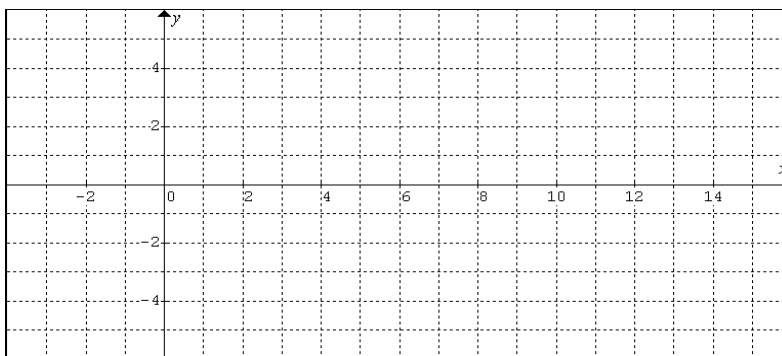
Sunset times at Parry Sound on these dates in 2001 are given in the table.

Date	Day of the year	Sunset time (standard time)
June 21	172	20.18 h
December 21	355	16.68 h

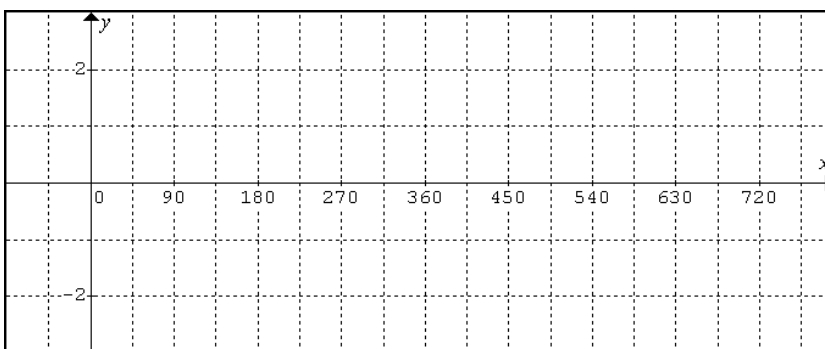
Determine an equation for the time of the sunset at Parry Sound on the n th day of the year. Show this information on a rough sketch.

Unit 6 Review – Sinusoidal Functions

1. Sketch the graph of a periodic function with a period of 6 and a range of 4 units on the grid below: (Yes, make up your own!)



2. Sketch both functions on the grid below; be sure to label your graphs:
 $y = 2 \cos x$ and $y = \sin 2x$

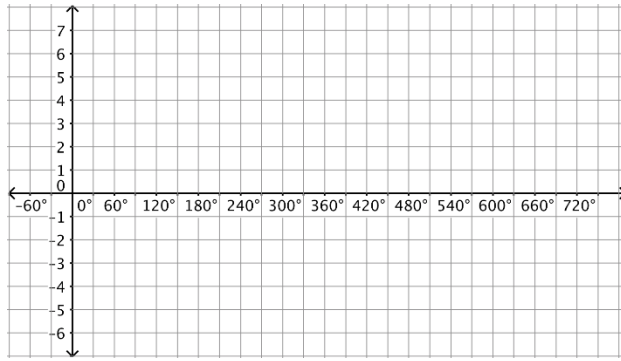


3. Write the equation of a sine function that has an amplitude of 3, and vertical translation of 5 units, and a period of 720° :

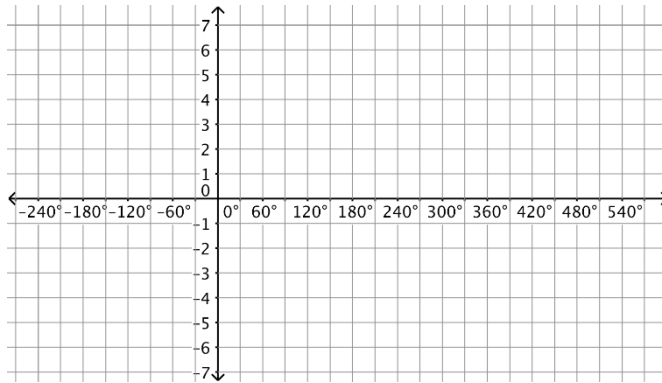
4. What is the phase shift of $y = 3 \sin 2(x - 135^\circ) + 1$: _____

5. Draw the following graphs on the grids provided.

a) $y = 3\sin 2x + 1$



b) $y = \cos \frac{1}{2}(-x - 180^\circ) - 2$



6. Write an equation for the Cosine function given the following:

a) amplitude of 3, period of 120° , and reaches its maximum value of 5 when $x = 45^\circ$.

b)



7. Describe using appropriate mathematical language the transformations required to graph

$y = 4\sin \frac{4}{3}(x + 45^\circ) - 2$ from $y = \sin x$.

8. If $f(2) = 6$ and $f(5) = 12$ for **periodic** function $f(x)$ with a period of 7 units, describe 2 other points that will be on this function and justify how you know.
9. Jessica is on a Ferris wheel. She gets on at the bottom at a height of 1m. Exactly 60sec later she is at the maximum height which is 16m.
- Sketch a graph to represent her height over time.
 - Write out her height over time as an equation using cosine: _____
 - Write out her height over time as an equation using sine: _____
 - At 42 seconds, what height is Jessica above the ground?
(Substitute 42 into one of your equations and solve for height).
10. A weight is supported by a spring. The weight rests 50 cm above a tabletop. The weight is pulled down 25 cm and released at time $t = 0$. This creates a periodic up-and-down motion. It takes 1.6 s for the weight to return to the low position each time. Assume the relationship between the height of the weight and time is sinusoidal. Determine the equation(use a sine and cosine function).

11. Determine an equation for the time of the sunset at Parry Sound on the n th day of the year. The latest sunsets occur around June 21 (the 172nd day of the year) ,sunset time is 20.18 and the earliest ones occur around December 21 (the 355th day of the year), sunset time is 16.68h
12. A carnival Ferris wheel with a radius of 7 m makes one complete revolution every 16 s. The bottom of the wheel is 1.5 m above the ground .
- Draw a graph to show how a person's height above the ground varies with time for three revolutions, starting when the person gets onto the Ferris wheel at its lowest point.
 - Find an equation for the graph.
 - Predict how the graph and the equation will change if the Ferris wheel turns more slowly.
13. The water depth in a harbor is 21 m at high tide and 11 m at low tide. One cycle is completed approximately every 12 h.
- Find an equation for the water depth as a function of the time, t hours, after low tide.
 - Draw a graph of the function for 48 h after low tide, which occurred at 14:00.